are retained near the top and others come down to various distances in the column.

Thin layer chromatography: This method is an another type of adsorption chromatography with this method it is possible to separate even minute quantities of mixtures. A sheet of a glass is coated with a thin layer of adsorbent (cellulose, silica gel or alumina). This sheet of glass is called chromoplate or thin layer chromatography plate. After drying the plate, a drop of the mixture is placed just above one edge and the plate is then placed in a closed jar containing eluent (solvent). The eluent is drawn up the adsorbent layer by capillary action. The components of the mixture move up along with the eluent to different distances depending upon their degree of adsorption of each component of the mixture. It is expressed in terms of its retention factor (ie) R_f value

$$R_f = \frac{\text{Distance moved by the substance from base line (x)}}{\text{Distance moved by the solvent from base line (y)}}$$

The spots of colored compounds are visible on TLC plate due to their original color. The colorless compounds are viewed under uv light or in another method using iodine crystals or by using appropriate reagent.

Partition chromatography: Paper chromatography (PC) is an example of partition chromatography. The same procedure is followed as in thin layer chromatography except that a strip of paper acts as an adsorbent. This method involves continues differential portioning of components of a mixture between stationary and mobile phase. In paper chromatography, a special quality paper known as chromatography paper is used. This paper act as a stationary phase.

A strip of chromatographic paper spotted at the base with the solution of the mixture is suspended in a suitable solvent which act as the mobile phase. The solvent rises up and flows over the spot. The paper selectively retains different components according to their different partition in the two phases where a chromatogram is developed. The spots of the separated colored compounds are visible at different heights from the position of initial spots on the chromatogram. The spots of the separated colorless compounds may be observed either under ultraviolent light or by the use of an appropriate spray reagent.

Evaluation

I. Choose the best answer.

1. Select the molecule which has only one π bond.

   a) $CH_3 - CH = CH - CH_3$

   b) $CH_3 - CH = CH - CHO$

   c) $CH_3 - CH = CH - COOH$

   d) All of these

2. In the hydrocarbon $\overset{7}{C}H_3 - \overset{6}{C}H_2 - \overset{5}{C}H = \overset{4}{C}H - \overset{3}{C}H_2 - \overset{2}{C} \equiv \overset{1}{C}H$ the state of hybridisation of carbon 1,2,3,4 and 7 are in the following sequence.

   a) sp, sp, sp^3, sp^2, sp^3

   b) sp^2, sp, sp^3, sp^2, sp^3

   c) sp, sp, sp^2, sp, sp^3

   d) none of these

3. The general formula for alkadiene is

   a) C_nH_{2n} b) C_nH_{2n-1}

   c) C_nH_{2n-2} d) C_nH_{n-2}

Unit 11 revised - jagan.indd 153 22-01-2021 12:42:32